Fundamentals of Biostatistics (Lectures 1-4)

BIOSTAT 201A Fall 2025

Discussion 2 – October X, 2025

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Outline

1. Lecture 1: Biostatistics, Inference, and Data
    1. Population vs. Sample, Parameter vs. Statistic
    2. Types of Data
2. Lecture 2 : Parameters vs. Statistics, Summary Statistics, and Data Visualization (See Problem 1)
    1. Measures of Location
    2. Measures of Spread
    3. Describing a Distribution
    4. Data Visualization - Boxplots
3. Lecture 3: Probability, Conditioning, and Independence (see Problems 2 and 3)
4. Lecture 4: Random Variables, Sampling, and The Central Limit Theorem
    1. Discrete Random Variables
    2. Properties of the Normal Distribution
    3. Central Limit Theorem

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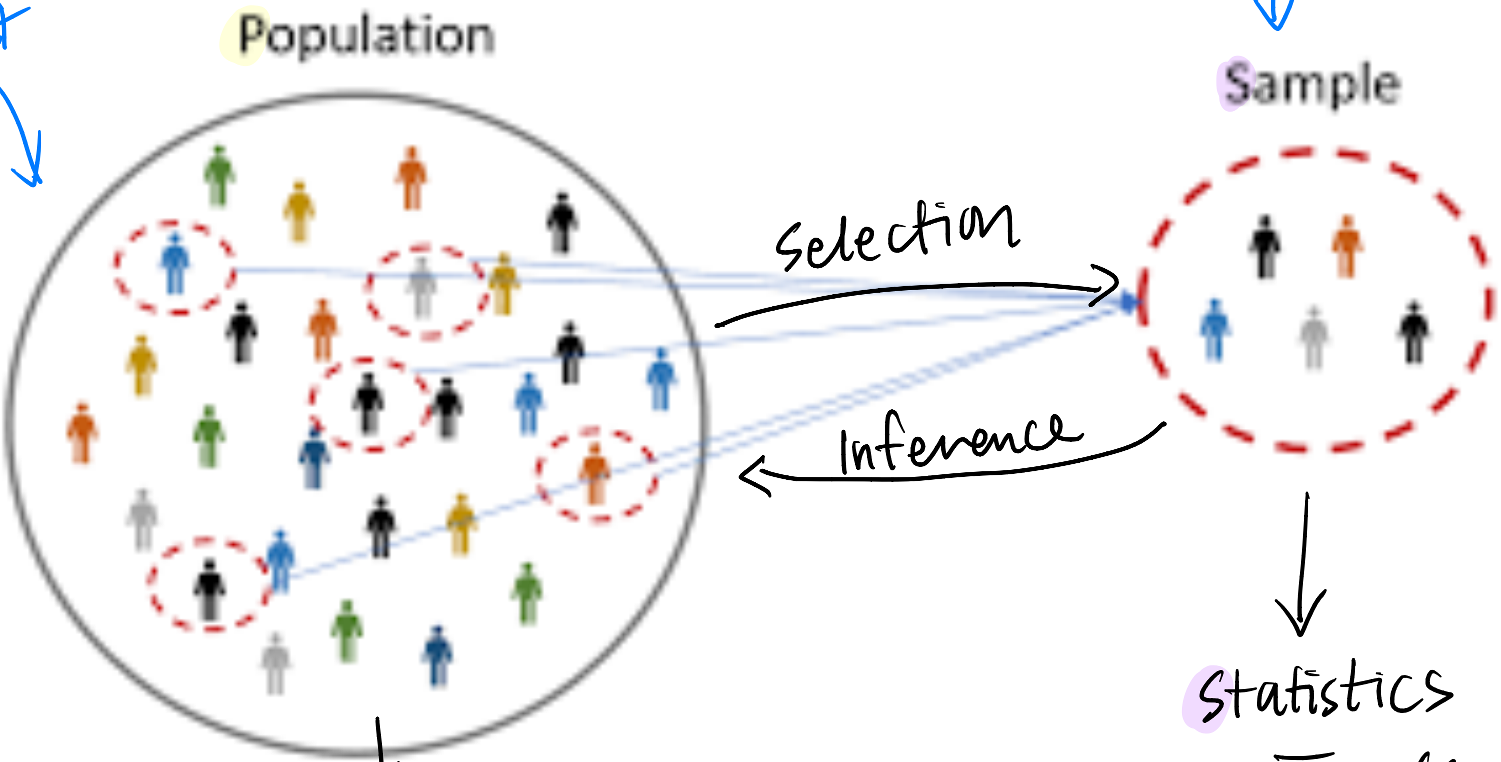
1.1 Population vs. Sample, Parameter vs. Statistic

what we want to learn about

what we have



Parameters:

μ (Pop'n Mean)

σ^2 (Pop'n Variance)

Statistics

$\bar{X}$ (Sample Mean)

s^2 (Sample Variance)

1.2 Types of Data

(1) **Quantitative (“Quantifiable”)** – discrete or continuous response of measurable magnitude

- Continuous – can take on an uncountable set of values
- Discrete – can take on a limited, and often fixed number of possible values

(2) **Qualitative –** no corresponding set of numbers

Can you produce some examples of each type of data?

1.2 Types of Data

- Stevens' Scale of Data Classification

	Categorical	Order	Equal Spacing	True Zero	Examples
Nominal – categorical data without any sensible ordering of values	X				voting party, race
Ordinal – categorical or discrete data with a natural ranking/ordering relationship	X	X			pain scale, letter grades
Interval – has a notion of distance, defined by measurement units of equal size	X	X	X		time on a clock, temperature (°F, °C)
Ratio – has a notion of magnitudes or "x times greater"	X	X	X	X	temperature (Kelvin) where 0 = absence of energy

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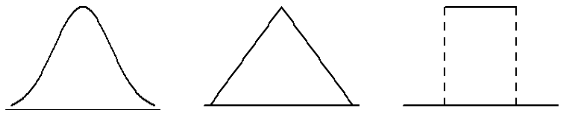
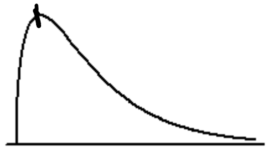
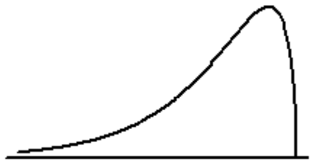
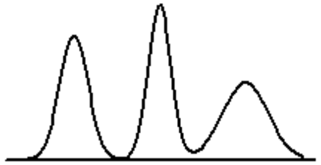
2.1 Measures of Location

	Formula/How to Calculate it	What the Data Looks Like
Mean	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$	 (a) Bell shaped (b) Triangular (c) Uniform (retangular)
Median	Let observations $x_1, \dots, x_n$ be ordered such that $x_1 \leq x_2 \leq \cdots \leq x_n$ where x_1 is the minimum and x_n is the maximum If n is even, the median is the average of $x_{\frac{n}{2}}$ and $x_{\frac{n+1}{2}}$ If n is odd, the median is $x_{\frac{n+1}{2}}$	 (d) Right skewed median < mean  (e) Left skewed
Mode	Most common value	 (i) Multi-modal

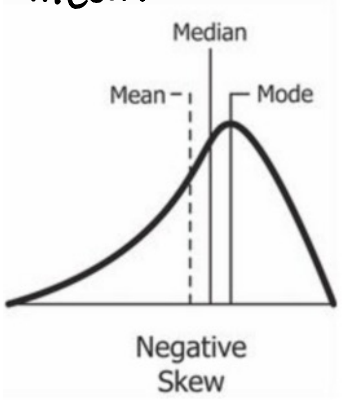
2.2 Measures of Spread

	Formula/How to Calculate it
Range	Range = $x_{max} - x_{min}$
Interquartile Range (IQR)	IQR = $x_{75th\ percentile} - \ x_{25th\ percentile} = Q3\ - Q1$
Sample Variance/Standard deviation ($\sqrt{s^2}$)	$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$

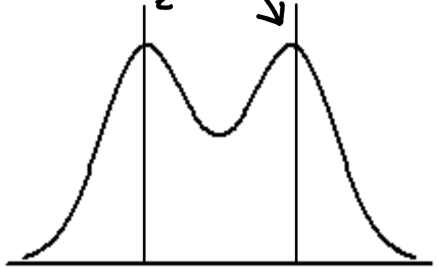
2.3 Describing a Distribution

Generally,
median for skewed distr.
mean for symmetric distr
mode for multiple humps

mean < median

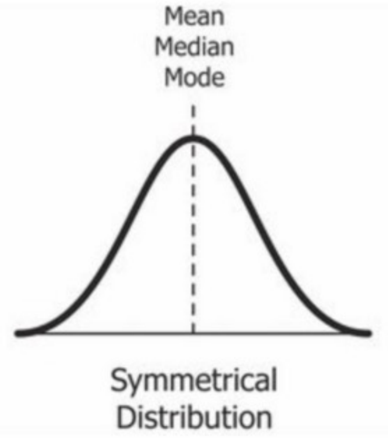


Modes

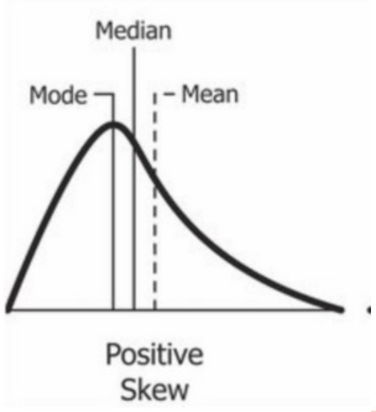


(h) Bimodal

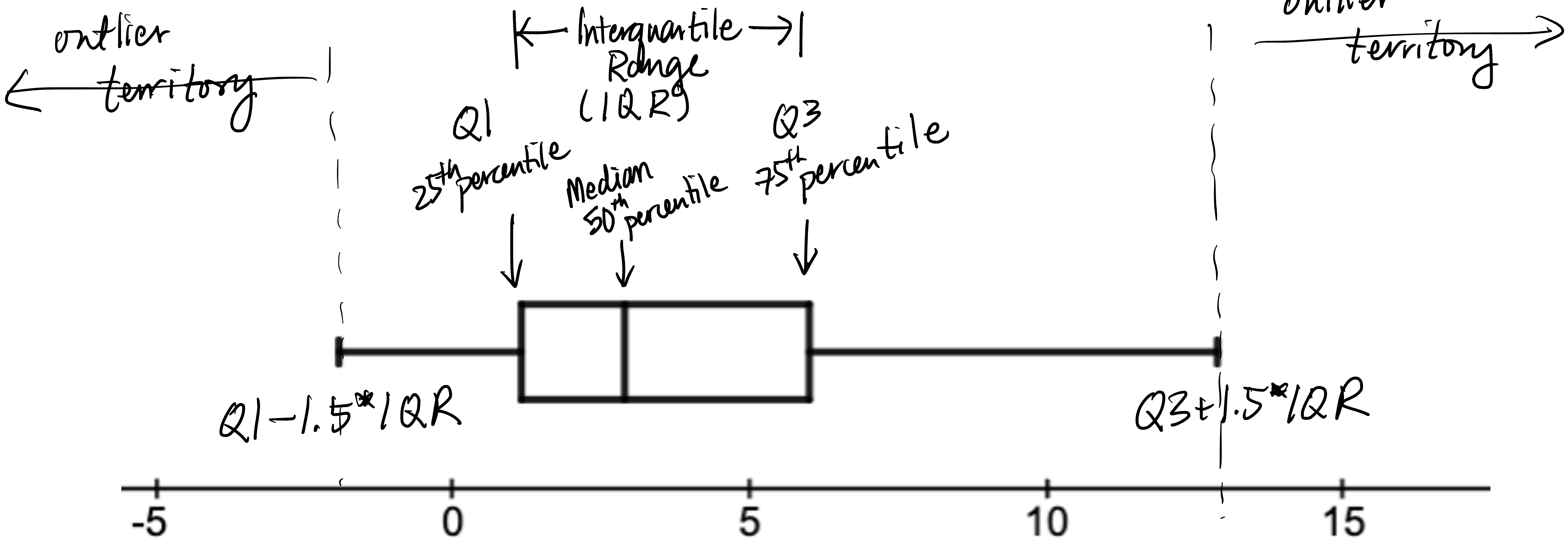
Mean ≈ median



mean > median



2.4 Data Visualization - Boxplots



2.4 Data Visualization – Stem and Leaf

15, 16, 21, 23, 23, 31, 35, 39, 40, 41, 46, 55, 56

1	5 6
2	1 3 3
3	1 5 9
4	0 1 6
5	5 6

Symmetric distr.

Problem 1. Summary Statistics and Data Visualization (Lecture 2)

The data in `serum_cholestrol.xlsx` are a sample of cholesterol levels taken from 24 hospital employees who were on a standard American diet and who agreed to adopt a vegetarian diet for 1 month. Serum-cholesterol measurements were made before adopting the diet and 1 month after. See Table 1 for the first 10 observations.

(Problem adopted from Exercises 2.13 - 2.18 Bernard Rosner, Fundamentals of Biostatistics, 2nd Edition).

(a) With your favorite statistical software, create a new column named `difference`, which is the difference between `Before` and `After` variables such that `difference = Before - After`.

(b) Generate the following summary statistics: mean, median, standard deviation, range, and IQR and comment on the distribution of the data. If you had to pick summary statistics to an investigator, which summary statistics would you choose and why?

Right skew
→ Median, IQR, sd.

mean = 19.542, Median = 19, Range = 62, IQR = 21.5 →

(c) Construct a stem-and-leaf plot of the cholesterol changes. (You can do this in Rguroo)

```
-0 | 308
 0 | 288233699
 2 | 13781256
 4 | 189
```

(d) Construct a box-plot of the cholesterol changes. Are there any outliers in your boxplot? Explain.

$Q1 = 10$, $\text{Median} = 19$, $Q3 = 31.5$

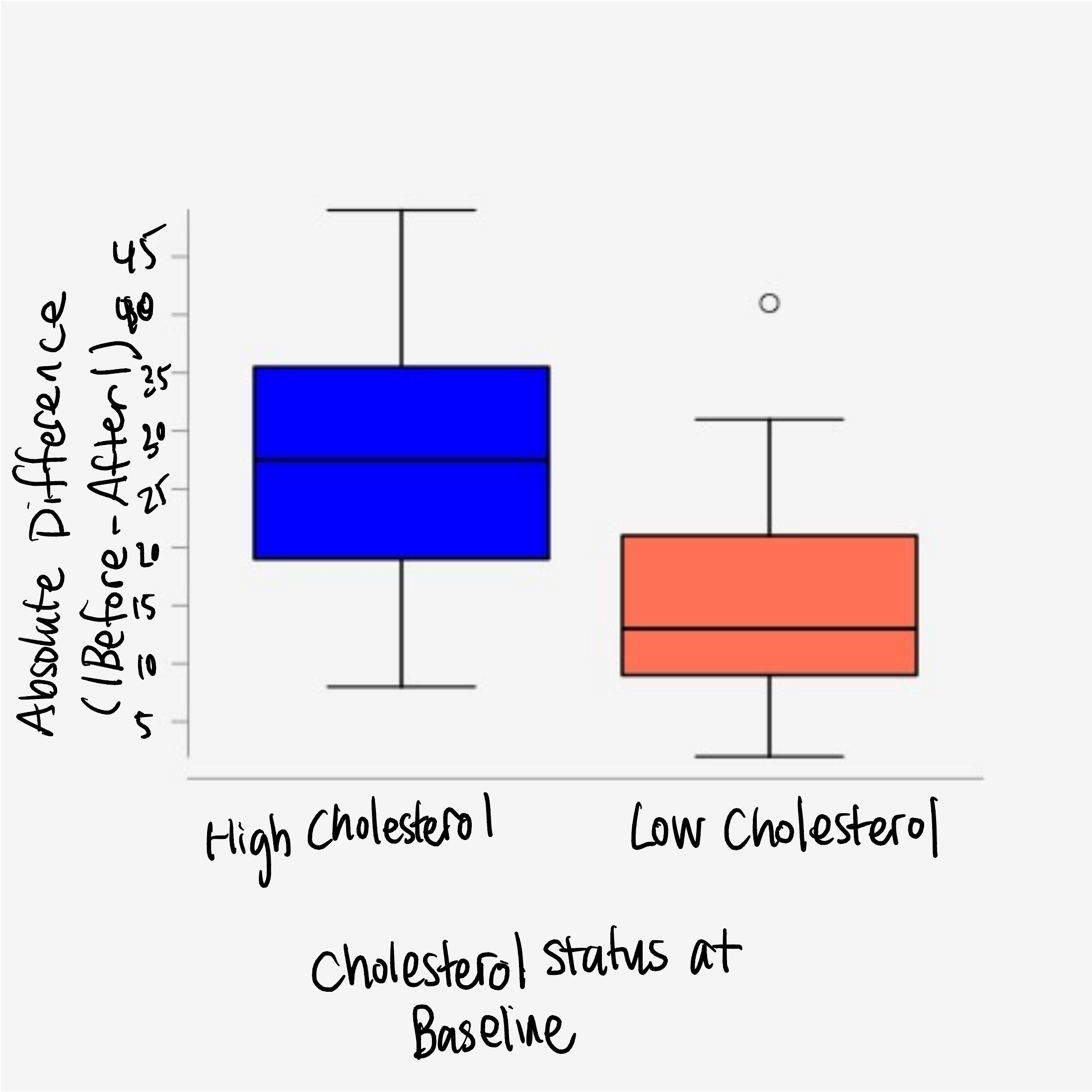
$Q3 + 1.5 * IQR = 63.75$
$Q1 - 1.5 * IQR = -22.25$
$\downarrow$
use min & max as whiskers if there are no outliers.

There are no outliers since the $\{\min, \max\} \in \{Q1 - 1.5 * IQR, Q3 + 1.5 * IQR\}$

(e) Some investigators believe that the effects of diet on cholesterol are more evident in people with high rather than low cholesterol levels. If you split the data in according to whether baseline cholesterol is above or below the median, can you comment descriptively on this issue? use absolute value to quantify difference

Hint: the best way to do this would be to create an indicator variable that indicates whether baseline cholesterol is above or below the median and then run the analysis.

$\text{Median}_{\text{Before}} = 179$



Absolute Difference
(|Before - After|)
45
40
35
30
25
20
15
10
5
High Cholesterol
Low Cholesterol
Cholesterol status at Baseline

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Some general probability rules

- Complements: $\mathrm{P}(\overline{\mathrm{E}}) = 1 - \mathrm{P}(\mathrm{E})$
- Independence of Events:
  (1) Event A is independent of Event B if the joint probability of A and B can be expressed as:
  $$\mathrm{P}(\mathrm{A} \cap \mathrm{B}) = \mathrm{P}(\mathrm{A})\mathrm{P}(\mathrm{B})$$
  (2) Event A is independent of Event B is the following conditional holds:
  $$\mathrm{P}(\mathrm{A}|\mathrm{B}) = \mathrm{P}(\mathrm{A}),\ \mathrm{P}(\mathrm{B}) > 0$$
- Conditional Probability: $\mathrm{P}(\mathrm{A}|\mathrm{B}) = \frac{\mathrm{P}(\mathrm{A}\cap\mathrm{B})}{\mathrm{P}(\mathrm{B})} = \frac{\mathrm{P}(\mathrm{A}|B)P(B)}{\mathrm{P}(\mathrm{B})}$
- Rule of Total Probability: $\mathrm{P}(\mathrm{A}) = \mathrm{P}(\mathrm{A}|\mathrm{B})\mathrm{P}(\mathrm{B}) + \mathrm{P}(\mathrm{A}|\overline{\mathrm{B}})\mathrm{P}(\mathrm{B})$

Problem 2. Dice Race (Lecture 3).

You and your friend are in a Dice Race where roll a 6-sided dice with numbers 1 through 6. You win the dice race if your number if your number is greater than your friends' number. For example, if I roll a 6 and my friend rolls a 2, I win the dice race.

(a) What is the probability that you win the dice race in one round?

You Roll

I Roll	1	2	3	4	5	6
1		✓	✓	✓	✓	✓
2			✓	✓	✓	✓
3				✓	✓	✓
4					✓	✓
5						✓
6						

$$Pr(\text{You Win}) = \frac{\text{Total \# of outcomes when } \underline{\text{You}} \text{ Win}}{\text{Total \# of outcomes}}$$

$$= \frac{(5+4+3+2+1)}{36}$$

$$= \frac{15}{36}$$

$$= \boxed{\frac{5}{12}}$$

(b) What is the probability that you win at least one round out of five rounds?

$$\Pr(\text{Win at least once}) = 1 - \underbrace{\Pr(\text{Win no rounds})}_{\Pr(\text{Lose every round})} = \boxed{1-\left(\frac{7}{12}\right)^5}$$

$$\Pr(\text{Lose every round}) = \_\_\_ \; \_\_\_ \; \_\_\_ \; \_\_\_ \; \_\_\_ \quad \leftarrow \text{5 independent events}$$

$$= \Pr(\text{Lose})^5$$

$$= \left(\frac{7}{12}\right)^5$$

$$\Pr(\text{Lose}) = 1 - \Pr(\text{Win})$$

$$= 1 - 5/12$$

$$= 7/12$$

Problem 3. The Freshman Flu? or Norovirus? (Lecture 3).

You are a doctor at the Arthur Ashe Student Center and it is allegedly freshman flu season. You have been observing a high percentage of students coming to the clinic with a high fever, a symptom of both the Flu and Norovirus.

In the general student population, it has been believed that the prevalence of students with the flu is 40% and the prevalence of students with Norovirus is 60%.

If a student has the Flu, the probability of the student developing a fever is 85% and if a student has Norovirus, the probability of the student developing a fever is 25%.

(a) For a randomly selected student, what is the probability that this patient has a Fever?

Let $\left.\begin{array}{ll} Pr(Flu) = 0.40 & Pr(Fever \mid Flu) = 0.85 \\ Pr(Noro) = 0.60 & Pr(Fever \mid Noro) = 0.25 \end{array}\right\} \Rightarrow$ use Law of Total Probability

Recall: $Pr(A \cap B) = Pr(A \mid B) Pr(B)$

$$\begin{aligned} Pr(Fever) &= Pr(Fever \cap Flu) + Pr(Fever \cap Noro) \leftarrow \\ &= Pr(Fever \mid Flu) P(Flu) + Pr(Fever \mid Noro) Pr(Noro) \\ &= 0.85(0.40) + 0.25(0.60) \\ &= \boxed{0.49} \end{aligned}$$

(b) If a student has a fever, what is the probability that the student has the Flu? What is the probability that the student has Norovirus?

$$Pr(Flu \mid Fever) = \frac{Pr(Fever \cap Flu)}{Pr(Fever)} = \frac{0.85(0.4)}{0.49} = 0.694$$

$$Pr(Noro \mid Fever) = \frac{Pr(Fever \cap Noro)}{Pr(Fever)} = \frac{Pr(Fever) - Pr(Fever \cap Flu)}{Pr(Fever)}$$

$$= 1 - Pr(Flu \mid Fever)$$

$$= 1 - 0.694$$

$$= \boxed{0.306}$$

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4.1 Discrete Random Variables - Example

Probability Mass Function (PMF) – probability distribution for a discrete variable

Value (x_i)	1 ← $x_i = 1$	2	4	6
Probability $Pr(X=x_i)$	0.2	0.1	0.4	0.3

$Pr(X=1)$ $\quad$ $Pr(X=2)$ . . .

Expected Value of a Discrete RV:

$$\mathbb{E}(X) = \sum_{i=1}^{n} x_i \Pr(X = x_i)$$

Variance of a Discrete RV:

$$Var(X) = \mathbb{E}(X - \mu)^2 = \sum_{i=1}^{n} x_i^2 \Pr(X = x_i) - \mu^2$$

Exercise. Compute

(a) Expected Value of X:

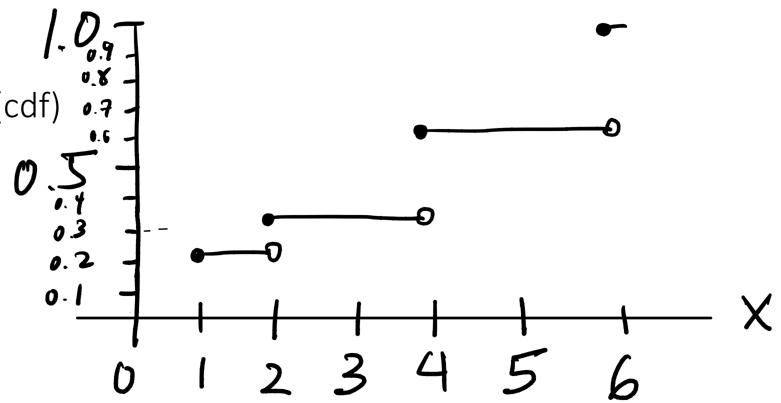
$$\mathbb{E}(X) = 1(0.2) + 2(0.1) + 4(0.4) + 6(0.3)$$
$$= 0.2 + 0.2 + 1.6 + 1.8 = \boxed{3.8}$$

(b) Variance of X:

$$Var(X) = 1^2(0.2) + 2^2(0.1) + 4^2(0.4) + 6^2(0.3) - 3.8^2 =$$

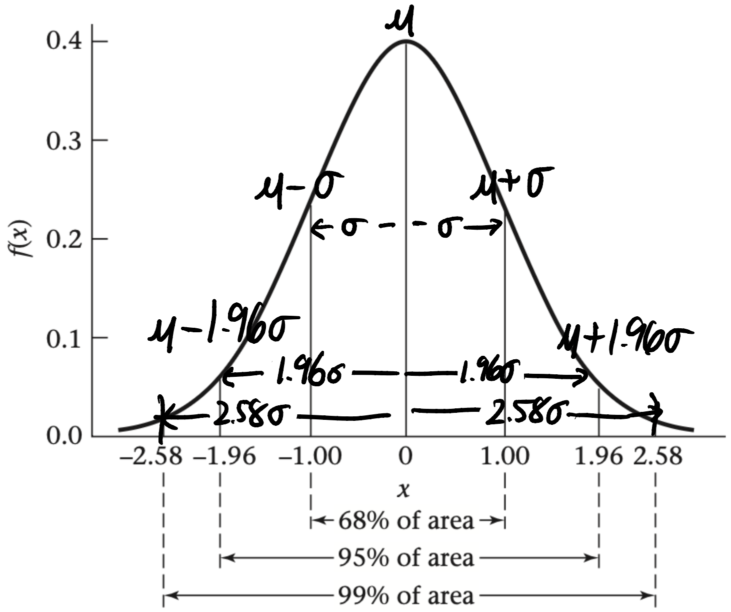
(c) Construct the cumulative distribution function (cdf)

x_i	1	2	4	6
$Pr(X=x_i)$	0.2	0.1	0.4	0.3
$Pr(X \leq x)$	0.2	0.3	0.7	1.0

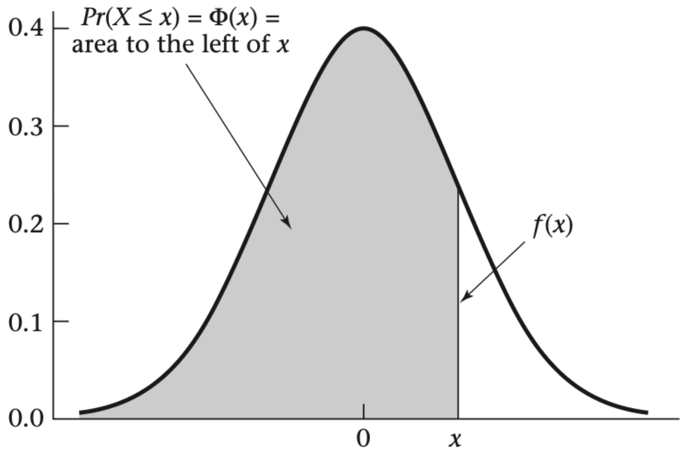


4.2 Properties of the Normal Distribution – The Empirical Rule and CDF

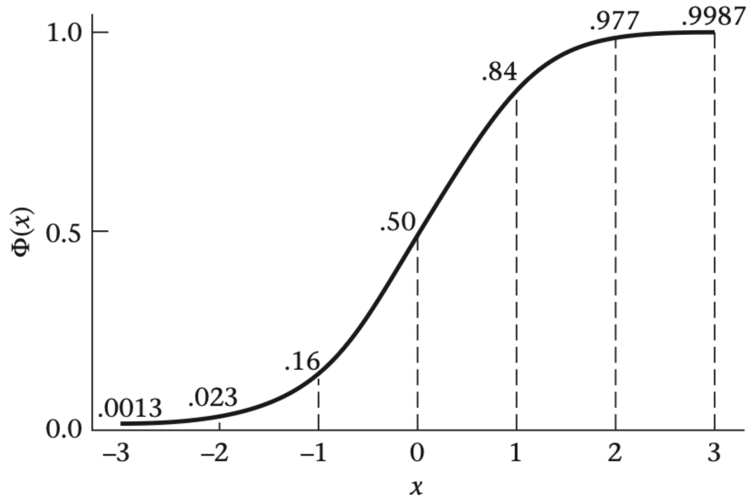
Empirical properties of the standard normal distribution



The cdf [Φ(x)] for a standard normal distribution



The cdf for a standard normal distribution [Φ(x)]



<u>Empirical Rule:</u>

68% of the data falls within 1 sd of the mean s.t. $Pr(\mu-\sigma < X < \mu+\sigma) = 68\%$

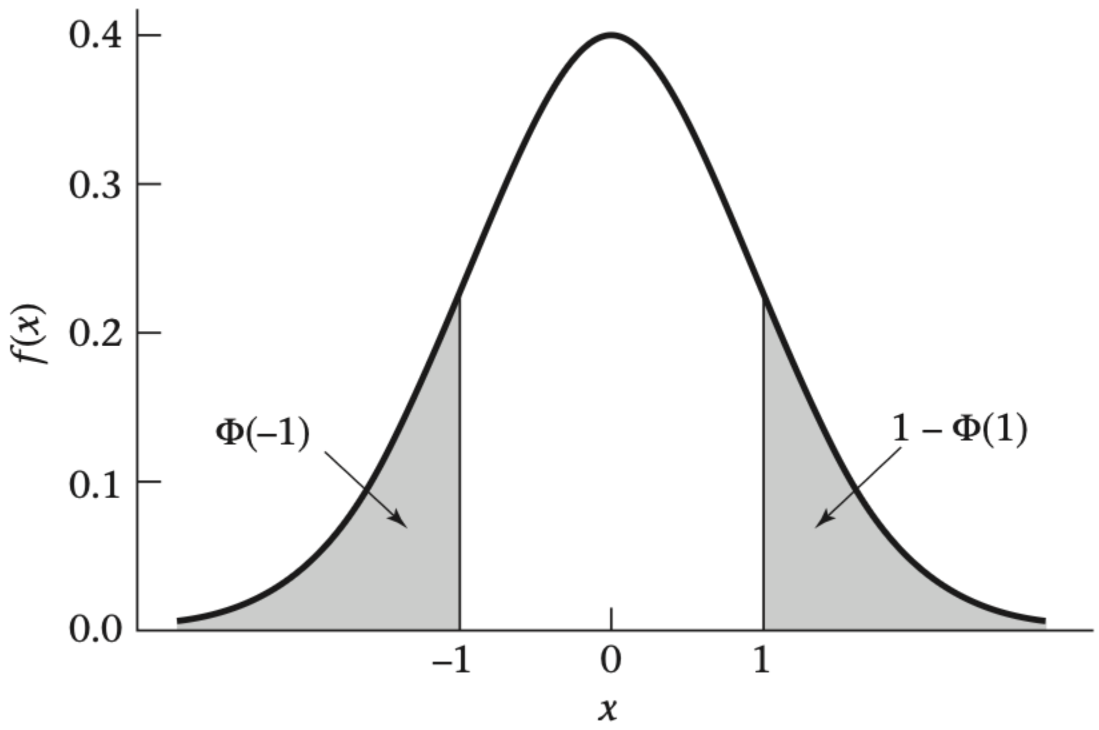
95% " " " ~2 sd " s.t. $Pr(\mu-1.96\sigma < X < \mu+1.96\sigma) = 95\%$

99% " " " ~3sd " s.t. $Pr(\mu-2.58\sigma < X < \mu+2.58\sigma) = 99\%$

4.2 Properties of the Normal Distribution - Symmetry

$$\Phi(-x) = 1 - \Pr(X \leq x) = 1 - \Phi(\mathrm{x})$$

Illustration of the symmetry properties of the normal distribution



4.3 Central Limit Theorem

draw $\overline{X}_1, \overline{X}_2, \ldots, \overline{X}_n$

$X \sim$ any distr.

where $\overline{X} = \{\overline{X}_1, \ldots, \overline{X}_n\} \sim N(\mu, \frac{\sigma^2}{n})$